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Transfer device and image forming apparatus

Abstract

A transfer device includes an intermediate transferor to rotate, primary transfer sections, a tension roller, a first movement mechanism, and a second movement mechanism. Each of the primary transfer sections includes a primary transferor. The tension roller stretches the intermediate transferor. The first movement mechanism causes the tension roller to move and change a position at which the tension roller stretches the intermediate transferor. The second movement mechanism causes the primary transferor of a primary transfer section to move to a contact position at which the primary transferor contacts a latent image bearer and a separation position at which the primary transferor is separated from the latent image bearer. The most-downstream primary transferor is movable between a contact position and a separation position. The first movement mechanism causes the tension roller to move to at least three positions at each of which the tension roller stretches the intermediate transferor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2022-091528, filed on Jun. 6, 2022, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

(2) Embodiments of the present disclosure relate to a transfer device and an image forming apparatus.

Related Art

(3) An image forming apparatus that prints a color image typically includes a transfer device for transferring toner of a special color such as a transparent color or a white color in addition to four colors of yellow (Y), magenta (M), cyan (C), and black (K). In such an image forming apparatus, first, toner images of the multiple colors are transferred to an intermediate transferor at primary transfer sections. Then, a multi-color toner image is secondarily transferred to a recording sheet such as a sheet of paper by a secondary transfer section.

(4) For example, in addition to the four colors of YMCK, a primary transfer section that transfers a toner image of a transparent color is disposed most downstream on an intermediate transfer belt in a rotation direction of the intermediate transfer belt. When the toner image of the transparent color is not formed, a primary transfer roller of the primary transfer section corresponding to the transparent color is separated from a photoconductor, and a toner image forming device of the transparent toner is stopped. As described above, the transfer roller that does not form the toner image of the special color is separated from the photoconductor that serves as a latent image bearer. Due to such a configuration, excessive consumption of the toner of the special color can be prevented.

SUMMARY

(5) In an embodiment of the present disclosure, a transfer device includes an intermediate transferor to rotate, multiple primary transfer sections, a tension roller, a first movement mechanism, and a second movement mechanism. The multiple primary transfer sections transfer developer images to the intermediate transferor and each of the plurality of primary transfer sections includes a primary transferor. The tension roller is disposed downstream from a most-downstream primary transferor of a most-downstream primary transfer section most downstream among the plurality of primary transfer sections in a rotation direction of the intermediate transferor, to stretch the intermediate transferor. The first movement mechanism causes the tension roller to move and change a position at which the tension roller stretches the intermediate transferor. The second movement mechanism causes the primary transferor of a primary transfer section upstream from the most-downstream primary transfer section in the rotation direction of the intermediate transferor to move to a contact position at which the primary transferor contacts a latent image bearer with the intermediate transferor interposed between the primary transferor and the latent image bearer and a separation position at which the primary transferor is separated from the latent image bearer. The most-downstream primary transferor is movable between a contact position at which the most-downstream primary transferor contacts another latent image bearer with the intermediate transferor interposed between the most-downstream primary transferor and

said another latent image bearer and a separation position at which the most-downstream primary transferor is separated from said another latent image bearer. The first movement mechanism causes the tension roller to move to at least three positions at each of which the tension roller stretches the intermediate transferor.

(6) In another embodiment of the present disclosure, an image forming apparatus includes the transfer device and multiple latent image bearers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more complete appreciation of the disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

(2) FIG. 1 is a schematic diagram illustrating a configuration of an image forming apparatus according to an embodiment of the present disclosure;

(3) FIG. 2 is a schematic diagram illustrating a configuration of a transfer device according to an embodiment of the present disclosure;

(4) FIGS. 3A, 3B, 3C, 3D, 3E, and 3F are diagrams illustrating the transfer device of FIG. 2 that operates in modes A, B, C, D, E and F, respectively, according to an embodiment of the present disclosure;

(5) FIG. 4 is a diagram illustrating how each of the modes A, B, C, D, E and F is switched between each other, according to an embodiment of the present disclosure;

(6) FIG. 5 is a perspective view of a driving source of a contact-and-separation mechanism as viewed from the front side of the image forming apparatus of FIG. 1, in which a primary transfer roller is arranged at a small separation position, according to an embodiment of the present disclosure;

(7) FIG. 6 is a perspective view of the driving source of the contact-and-separation mechanism of FIG. 5, in which a bracket covering a gear train is removed, according to an embodiment of the present disclosure;

(8) FIG. 7 is a perspective view of a driving source of a contact-and-separation mechanism as viewed from the front side of the image forming apparatus of FIG. 1, in which a primary transfer roller is arranged at a contact position, according to an embodiment of the present disclosure;

(9) FIG. 8 is a perspective view of the driving source of the contact-and-separation mechanism of FIG. 5, viewed from the front side of the image forming apparatus of FIG. 1, in which a primary transfer roller is arranged at a large separation position, according to an embodiment of the present disclosure;

(10) FIG. 9 is a cross-sectional view of a contact-and-separation mechanism viewed from the back side of the image forming apparatus of FIG. 1, in which a primary transfer roller of a most-downstream primary transfer section is arranged at a contact position relative to an intermediate transfer belt, according to an embodiment of the present disclosure;

(11) FIG. 10 is a cross-sectional view of the contact-and-separation mechanism of FIG. 9, in which the primary transfer roller of the most-downstream primary transfer section is arranged at a small separation position relative to the intermediate transfer belt, according to an embodiment of the present disclosure;

(12) FIG. 11 is a cross-sectional view of the contact-and-separation mechanism of FIG. 9, in which the primary transfer roller of the most-downstream primary transfer section is arranged at a large separation position relative to the intermediate transfer belt, according to an embodiment of the present disclosure;

(13) FIG. 12 is a perspective view of a cam according to an embodiment of the present disclosure;

(14) FIG. 13 is a perspective view of a cam and components around the cam viewed from the back side of FIG. 12, according to an embodiment of the present disclosure;

(15) FIG. 14 is a plan view of a configuration around a first arm and a second arm according to an embodiment of the present disclosure;

(16) FIG. 15 is a perspective view of the second arm of FIG. 14 and components around the second arm, according to an embodiment of the present disclosure;

(17) FIG. 16 is a perspective view of the second arm of FIG. 14 and components around the second arm viewed from the back side of FIG. 15, according to an embodiment of the present disclosure;

(18) FIG. 17 is a plan view of a configuration around a first sensor bracket and a sensor, according to an embodiment of the present disclosure;

(19) FIG. 18 is a perspective view of a first sensor bracket and a second sensor bracket as viewed from the front side of the image forming apparatus of FIG. 1, according to an embodiment of the present disclosure;

(20) FIG. 19 is a plan view of a second sensor bracket and a rotator in which the second sensor bracket is arranged when the primary transfer roller is arranged at the large separation position, according to an embodiment of the present disclosure;

(21) FIG. 20 is a side view of a configuration in which a central primary transfer section and a most-upstream primary transfer section contact with or separate from an intermediate transfer belt, according to an embodiment of the present disclosure;

(22) FIG. 21 is a diagram illustrating an arrangement of image forming devices, pre-supply reservoirs, and toner bottles in a case in which a toner bottle of a special color is arranged in a most-downstream primary transfer section, according to an embodiment of the present disclosure;

(23) FIG. 22 is a diagram illustrating an arrangement of the image forming devices, the pre-supply reservoirs, and the toner bottles of FIG. 21 in a case in which a toner bottle for black toner is arranged in a most-downstream primary transfer section, according to an embodiment of the present disclosure;

(24) FIG. 23 is a schematic diagram illustrating a configuration of a toner supply device according to an embodiment of the present disclosure;

(25) FIG. 24 is a flowchart of a process for checking arrangement of image forming devices, pre-supply reservoirs, and toner bottles, according to an embodiment of the present disclosure;

(26) FIG. 25 is a schematic diagram illustrating a configuration of a controller disposed in the image forming apparatus of FIG. 1, according to an embodiment of the present disclosure;

(27) FIG. 26 is a diagram illustrating an arrangement of driven rollers and a sensor, according to an embodiment different from the embodiment of FIG. 5;

(28) FIGS. 27A and 27B are side views of a configuration in which primary transfer rollers of a central primary transfer section contact with or separate from an intermediate transfer belt, according to a modification of the embodiment of FIG. 5; FIG. 27A is a plan view of the central primary transfer section in which the primary transfer rollers of the central primary transfer section contacts an intermediate transfer belt, according to the modification;

(29) FIGS. 28A, 28B, and 28C are plan views of a configuration in which a primary transfer roller of a most-downstream primary transfer section according to a modification of the embodiment of FIG. 5 different from the modification of FIGS. 27A and 27B;

(30) FIG. 28A is a plan view of the most-downstream primary transfer section in which the primary transfer roller of the most-downstream primary transfer section is arranged at a contact position;

(31) FIG. 28B is a plan view of the most-downstream primary transfer section in which the primary transfer roller of the most-downstream primary transfer section is arranged at a small separation position;

(32) FIG. 28C is a plan view of the most-downstream primary transfer section in which the primary transfer roller of the most-downstream primary transfer section is arranged at a large separation position;

(33) FIG. 29 is a schematic diagram illustrating a configuration of a transfer device according to a modification of the embodiment of FIG. 5 different from the modification of FIGS. 27A and 27B in which a primary transfer roller of a most-downstream primary transfer section is arranged at a contact position;

(34) FIGS. 30A and 30B are plan views of a rotation mechanism for rotating a tension roller, according to an embodiment of the present disclosure;

(35) FIG. 30A is a diagram illustrating the rotation mechanism in which a primary transfer roller of a most-downstream primary transfer section is arranged at a contact position; and

(36) FIG. 30B is a diagram illustrating the rotation mechanism in which the primary transfer roller of the most-downstream primary transfer section is arranged at a separation position.

(37) The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

(38) In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

(39) Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(40) Embodiments of the present disclosure are described below with reference to the drawings in the following description. In the drawings, like reference signs denote like or equivalent components and overlapping description of those components may be simplified or omitted as appropriate.

(41) FIG. 1 is a diagram illustrating a configuration of an image forming apparatus 1 according to an embodiment of the present disclosure. The image forming apparatus 1 illustrated in FIG. 1 is a tandem-type color printer in which multiple photoconductors as latent image bearers are arranged in parallel. Each of the photoconductors provided for the image forming apparatus 1 can form a toner image in a color corresponding to a color separation component of a color image using toner as developer supplied from a developing device. After the toner images formed on the photoconductors are superimposed and transferred to an intermediate transferor, the superimposed images are collectively transferred to a sheet such as a recording sheet. By so doing, a multicolor image can be formed on the sheet. In embodiments of the present disclosure, the image forming apparatus 1 is not limited to a color printer. However, no limitation is indicated thereby, and the image forming apparatus 1 may be, for example, a color copier, a facsimile apparatus, or a printing machine.

(42) As illustrated in FIG. 1, the image forming apparatus 1 includes an image former 1A in a center portion of the image forming apparatus 1 in the vertical direction, a sheet feeder 1B below the image former 1A, and a document scanner 1C including a document loading table 1C1 above the image former 1A. The image former 1A includes an intermediate transfer belt 2 as an intermediate transferor. The intermediate transfer belt 2 has a stretched surface in a horizontal direction. The image forming apparatus 1 includes components that form images in colors complementary to color separation colors above the intermediate transfer belt 2.

(43) In the image former 1A, image forming devices 10K, 10C, 10M, 10Y, and 10T are arranged. The image forming devices 10K, 10C, 10M, and 10Y can form images with toners of colors of yellow, magenta, cyan, and black, respectively, in a complementary color relation. The image forming device 10T forms a glossy image with transparent toner. In each of the multiple image

forming devices **10K**, **10C**, **10M**, **10Y**, and **10T**, photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T**, respectively, that can bear images are arranged in parallel along the stretched surface of the intermediate transfer belt **2**. The photoconductor **3T** bears an image of a transparent toner. In the following description, each of the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** may be simply referred to as a photoconductor **3** in a case in which a similar description applies to all the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T**.

(44) Each of the multiple photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** is made of a drum rotatable in the same direction, which is a counterclockwise direction in FIG. **1**. Around each of the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T**, a charger, a writing device, a developing device **6**, a primary transfer roller as a primary transfer section, and a cleaner are arranged. Each of the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T**, the charger, the writing device, the developing device **6**, the primary transfer roller **7**, and the cleaner collectively perform image forming processing when the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** rotate. For the sake of convenience, a developing device **6T** and a primary transfer roller **7T** provided for the photoconductor **3T** includes the reference sign **T**.

(45) A transfer device **20** includes the intermediate transfer belt **2**, primary transfer rollers **7K**, **7Y**, **7M**, **7C**, and **7T** (see FIG. **2**) as primary transferors, and rollers **2A** and **2B** and a secondary-transfer backup roller **2C**. Only the primary transfer roller **7T** is illustrated with a reference sign in FIG. **1** for the sake of convenience.

(46) Toner images formed in the image forming devices **10K**, **10C**, **10M**, **10Y**, and **10T** including the multiple photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T**, respectively, are sequentially transferred to the intermediate transfer belt **2**. The intermediate transfer belt **2** is stretched around the rollers **2A** and **2B**, the secondary-transfer backup roller **2C**, and multiple rollers that are not denoted with reference signs in FIG. **1**, to rotate in a direction indicated by arrow **A** in FIG. **1**. The intermediate transfer belt **2** faces the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** at multiple positions. The rollers **2A** and **2B** stretch the intermediate transfer belt **2** at two positions outer than the multiple positions in the direction of rotation of the intermediate transfer belt **2**. The secondary-transfer backup roller **2C** faces the secondary transfer device **9** with the intermediate transfer belt **2** interposed between the secondary-transfer backup roller **2C** and the secondary transfer device **9**.

(47) The secondary transfer device **9** includes a secondary transfer roller **9A**. The secondary transfer roller **9A** forms a secondary transfer nip at a position at which the secondary transfer roller **9A** presses against the secondary-transfer backup roller **2C** with the intermediate transfer belt **2** interposed between the secondary transfer roller **9A** and the secondary-transfer backup roller **2C**. A secondary transfer bias having the same polarity as the polarity of toner is applied to the secondary-transfer backup roller **2C**. On the other hand, the secondary transfer roller **9A** is grounded. Accordingly, a secondary transfer electric field is formed at the secondary transfer nip. The secondary transfer electric field electrostatically moves a multicolor toner image on the intermediate transfer belt **2** from the intermediate transfer belt **2** toward the secondary transfer roller **9A**. The secondary transfer device **9** transfers the multicolor toner image onto a sheet, which is conveyed to the secondary transfer nip at the secondary transfer nip.

(48) A recording sheet is fed to the secondary transfer nip from a sheet feeder **1B**. The sheet feeder **1B** includes multiple sheet feed trays **1B1** and multiple conveyance rollers **1B2**. The multiple conveyance rollers **1B2** are disposed on a conveyance path of recording sheets fed from the sheet feed trays **1B1**.

(49) The photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** are irradiated with writing light by the corresponding one of the writing devices **5**, and electrostatic latent images corresponding to image data are formed on the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T**. The image data is obtained by scanning a document on the document loading table **1C1** disposed in the document scanner **1C**, or by image data output from a computer.

(50) The document scanner **1C** includes a scanner **1C2** and an automatic document feeder **1C3**. The

scanner **1C2** exposes and scans a document on the document loading table **1C1**. The automatic document feeder **1C3** is disposed above an upper surface of the document loading table **1C1**. The automatic document feeder **1C3** inverts a document fed onto the document loading table **1C1** to scan front and back sides of the document.

(51) Each of the electrostatic latent images on the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** formed by the writing devices **5** is subjected to visual image processing by the corresponding one of the developing devices **6K**, **6C**, **6M**, **6Y**, and **6T** and primarily transferred to the intermediate transfer belt **2**. The developing device **6T** is illustrated in FIG. **1** for the sake of convenience. After toner images of black, yellow, cyan, magenta, and transparent colors are superimposed and transferred onto the intermediate transfer belt **2**, the toner images are secondarily transferred onto a recording sheet collectively by the secondary transfer device **9**.

(52) Subsequently, a multicolor image to be fixed bome on the surface of the recording sheet on which the secondary transfer has been performed is fixed by the fixing device **11**. The fixing device **11** has a belt fixing structure in which a fixing belt heated by a heating roller and a pressure roller facing and in contact with the fixing belt are disposed. In such a configuration, a contact area, in other words, a nip area is disposed between the fixing belt and the pressure roller, thus allowing an area in which the recording sheet is heated to be increased as compared with a heat-roller fixing structure.

(53) A conveyance direction of the recording sheet that has passed through the fixing device **11** can be switched by a conveyance-path switching claw disposed in a rear portion of the fixing device **11**. Specifically, the conveyance direction of the recording sheet is selected between the conveyance path directed to a sheet ejector **13** and a reverse conveyance path RP by the conveyance-path switching claw.

(54) In the image forming apparatus **1** having the above-described configuration, electrostatic latent images are formed on the uniformly charged photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** by exposure scanning of a document placed on the document loading table **1C1** or by reading image data from a computer. Subsequently, the electrostatic latent images are subjected to visual image processing by the developing devices **6K**, **6C**, **6M**, **6Y**, and **6T**. Then, the toner images are primarily transferred to the intermediate transfer belt **2**.

(55) In the case of a single-color image, a toner image that has been transferred to the intermediate transfer belt **2** is transferred onto a recording sheet fed from the sheet feeder **1B** as is. In the case of a multi-color image, primary transfer is repeated such that toner images are superimposed one on another. Then, the toner images are secondarily transferred to the recording sheet collectively. The unfixed image that has been secondarily transferred onto the recording sheet is fixed by the fixing device **11**. Then, the recording sheet is fed to the sheet ejector **13** or reversed and fed again to the secondary transfer nip.

(56) In FIG. **1**, the intermediate transfer belt **2** is formed of, for example, a single layer or multiple layers of polyvinylidene fluoride (PVDF), ethylene-tetrafluoroethylene copolymer (ETFE), polyimide (PI), or polycarbonate (PC). A conductive material such as carbon black is dispersed in the intermediate transfer belt **10**. The intermediate transfer belt **2** is adjusted to have a volume resistivity in a range of 10^8 to 10^{12} Ωcm and a surface resistivity in a range of 10^9 to 10^{13} Ωcm . The surface of the intermediate transfer belt **2** may be coated with a release layer as needed. Examples of the material employed for coating the intermediate transfer belt **2** include fluororesins such as ethylene-tetrafluoroethylene copolymer (ETFE), polytetrafluoroethylene (PTFE), polyvinylidene fluoride (PVDF), perfluoroalkoxy fluororesin (PFA), tetrafluoroethylene-hexafluoropropylene copolymer (FEP), and vinyl fluoride (PVF). However, the materials employed for coating the intermediate transfer belt **2** are not limited to the above-described fluororesins. Examples of a method for producing the intermediate transfer belt **2** include a casting method and a centrifugal molding method. The surface of the intermediate transfer belt **2** may be polished as needed. When the volume resistivity of the intermediate transfer belt **2** exceeds the above-described

range, a bias needed to transfer a toner image onto a recording sheet increases. Accordingly, the cost of power source for the intermediate transfer belt 2 is increased. For this reason, such a configuration of the intermediate transfer belt 2 is not preferable. Further, charging potential of the intermediate transfer belt 2 increases in, for example, a transfer process, or a transfer-sheet peeling process. Accordingly, self-discharge of the intermediate transfer belt 2 may be difficult. For this reason, an electric-charge remover is needed. In addition, when the volume-resistivity and the surface-resistivity of the intermediate transfer belt 2 are lower than the above-described ranges, attenuation of the charging potential is fast, which is advantageous for removing electric charges of the intermediate transfer belt 2 due to self-discharge. However, an electric current at the time of transfer flows in a plane direction of the surface of the intermediate transfer belt 2. Accordingly, toner scattering may occur. For this reason, the volume resistivity and the surface resistivity of the intermediate transfer belt 2 according to the present embodiment are preferably set within the ranges described above. Note that, for the measurement of the volume resistivity and the surface resistivity of the intermediate transfer belt 2, a high-resistance resistivity meter (Hiresta-IP, registered trademark, manufactured by Mitsubishi Chemical Corporation) was connected to a high resistance state (HRS) probe having the inner electrode diameter of 5.9 mm and the ring-electrode inner-diameter of 11 mm. A voltage of 100 V with the surface resistivity of 500 V was applied to the front and back surfaces of the intermediate transfer belt 2 and a measured value after 10 seconds from a time at which the voltage of 100 V and the surface resistivity of 500 V was applied, was employed.

(57) The intermediate transfer belt 2 is stretched around at least the roller 2A and the roller 2B as a roller pair and the secondary-transfer backup roller 2C disposed at the secondary transfer nip. The roller 2A as a driving roller is set to rotate clockwise such that the intermediate transfer belt 2 moves in the direction indicated by arrow A illustrated inside the intermediate transfer belt 2 in FIG. 1. The surface of the intermediate transfer belt 2, on which the toner images are transferred, moving between the roller 2A and the roller 2B faces the multiple photoconductors 3K, 3Y, 3C, 3M, and 3T of the image forming devices 10K, 10C, 10M, 10Y, and 10T. The primary transfer rollers 7K, 7Y, 7M, 7C, and 7T serve as transferors for electrostatically transferring visible images on the respective photoconductors 3 to the intermediate transfer belt 2. The primary transfer rollers 7K, 7Y, 7M, 7C, and 7T are disposed at positions at which the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T face the photoconductors 3K, 3C, 3M, 3Y, and 3T, respectively, via the intermediate transfer belt 2. The primary transfer roller 7T is illustrated in FIG. 1 for the sake of convenience.

(58) The primary transfer rollers 7K, 7Y, 7M, 7C, and 7T according to the present embodiment are cored bars made of metal such as iron, steel use stainless (SUS), or aluminum (Al) coated with foam resin. The foam resin has a wall thickness of 2 mm to 10 mm. Blade-shaped or brush-shaped transferors known in the art can also be employed as the transferors.

(59) In the present embodiment, white toner is employed for the purpose of forming a white base color for an image in addition to toner employed for full-color image formation. In addition, transparent toner may be employed for the purpose of improving glossiness and transferability of an image, and, for example, light cyan toner, or light magenta toner may be selected for increasing a color gamut. For the purpose of creating a colored metal color such as a red copper color and a bronze color, toner of a metal color such as gold toner and silver toner may also be employed as a base.

(60) As illustrated in FIG. 2, the primary transfer roller 7T and the photoconductor 3T form a special color transfer nip NT with the intermediate transfer belt 2 interposed between the primary transfer roller 7T and the photoconductor 3T. The primary transfer roller 7C and the photoconductor 3C form a cyan transfer nip NC with the intermediate transfer belt 2 interposed between the primary transfer roller 7C and the photoconductor 3C. The primary transfer roller 7M and the photoconductor 3M form a magenta transfer nip NM with the intermediate transfer belt 2 interposed between the primary transfer roller 7M and the photoconductor 3M. The primary

transfer roller 7Y and the photoconductor 3Y form a yellow transfer nip NY with the intermediate transfer belt 2 interposed between the primary transfer roller 7Y and the photoconductor 3Y. The primary transfer roller 7K and a photoconductor 3K form a black transfer nip NM with the intermediate transfer belt 2 interposed between the primary transfer roller 7K and the photoconductor 3K.

(61) The transfer device 20 includes a most-upstream primary transfer section 201 disposed most upstream in the rotation direction of the intermediate transfer belt 2, a most-downstream primary transfer section 203 disposed most downstream in the rotation direction of the intermediate transfer belt 2, and a central primary transfer section 202 including the primary transfer rollers 7Y, 7M, and 7C disposed between the most-upstream primary transfer section 201 and the most-downstream primary transfer section 203. In the present embodiment, the most-upstream primary transfer section 201 transfers a black toner image at a black transfer nip NK, the central primary transfer section 202 transfers a cyan toner image at a cyan transfer nip NC, a magenta toner image at a magenta transfer nip NM, and a yellow toner image at a yellow transfer nip NY to the intermediate transfer belt 2. The most-downstream primary transfer section 203 transfers a special color toner image at a special color transfer nip NT to the intermediate transfer belt 2. Furthermore, in the following description, upstream or downstream in the rotation direction of the intermediate transfer belt 2 may be also referred to simply as upstream or downstream.

(62) In FIG. 2, the primary transfer roller 7K disposed in the most-upstream primary transfer section 201 is a most-upstream primary transferor, the primary transfer rollers 7Y, 7M, and 7C disposed in the central primary transfer section 202 are central primary transferors, and the primary transfer roller 7T disposed in the most-downstream primary transfer section 203 is a most downstream primary transferor. The rotation direction of the intermediate transfer belt 2 is a direction indicated by arrow A in FIG. 2. The primary transfer rollers 7K, 7Y, 7M, and 7C upstream from the primary transfer roller 7T in the rotation direction of the intermediate transfer belt 2 are also upstream primary transferors.

(63) In the present embodiment, a toner image of the special color can be transferred to the intermediate transfer belt 2 in both the most-upstream primary transfer section 201 and the most-downstream primary transfer section 203. Accordingly, a toner image of the special color can be transferred in a desired order. Details are described below.

(64) Between the primary transfer roller 7C and the primary transfer roller 7T in the rotation direction of the intermediate transfer belt 2, a driven roller 21A as a second tension roller and a sensor 22 as a sensor are disposed. The driven roller 21A stretches the intermediate transfer belt 2. The sensor 22 detects a scale on the intermediate transfer belt 2 and detects the rotation speed of the intermediate transfer belt 2. Controlling the rotation speed of the intermediate transfer belt 2 based on the detection result of the sensor 22 prevents positional shift of toner images of the colors to be transferred to the intermediate transfer belt 2.

(65) In the transfer device 20 according to the present embodiment, the multiple primary transfer rollers 7K, 7Y, 7M, 7C, and 7T contact with and separate from the photoconductors 3K, 3Y, 3M, 3C, and 3T, respectively, with the intermediate transfer belt 2 interposed between the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T and the photoconductors 3K, 3Y, 3M, 3C, and 3T, respectively, in accordance with modes of image formation. Specifically, as described in modes A, B, C, D, E, and F in Table 1 given below, the position of each of the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T can be changed between a contact position and a separation position. The contact position is a position at which each of the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T contacts the corresponding one of the photoconductors 3K, 3Y, 3M, 3C, and 3T, via the intermediate transfer belt 2 to form a primary transfer nip. The separation position is a position at which each of the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T is separated from the corresponding one of the photoconductors 3K, 3Y, 3M, 3C, and 3T. The driven roller 21A around which the intermediate transfer belt 2 is stretched and the driven roller 33A that serves as a first

tension roller also move in a direction away from the photoconductor 3T in conjunction with the primary transfer roller 7T of the most-downstream primary transfer section 203, in other words, in a downward direction in FIG. 2 or in an upward direction opposite to the downward direction. The position of the primary transfer roller 7T of the most-downstream primary transfer section 203 can be changed among the following positions: the contact position at which the primary transfer roller 7T contacts the photoconductor 3T to form the primary transfer nip NT, a small separation position at which the primary transfer roller 7T is separated from the photoconductor 3T by a small separation distance, and a large separation position at which the primary transfer roller 7T is separated from the photoconductor 3T by a large separation distance. In conjunction with the primary transfer roller 7T, the driven rollers 21A and 33A also move in the upward direction in FIG. 2, which is a direction in which the driven rollers 21A and 33A approach the photoconductor 3T or in the downward direction in FIG. 2, which is a direction in which the driven rollers 21A and 33A move away from the photoconductor 3T. FIG. 2 illustrates a case of the mode D in which all the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T contact with the intermediate transfer belt 2.

(66) TABLE-US-00001

TABLE 1	A	B	C	D	E	F	Most-	Small	Contact	Contact	Contact	Large	Small
downstream	separation	position	position	position	separation	separation	primary	transfer	position				
position	position	section	+	Driven	roller	Central	primary	Separation	Separation	Contact	Contact		
Contact	Separation	transfer	section	position	position	position	position	position	position	position	position	Most-	
upstream	Separation	Separation	Separation	Contact	Contact	Contact	primary	transfer	position				
position	position	position	position	position	position	Section							

(67) FIGS. 3A, 3B, 3C, 3D, 3E, and 3F are diagrams illustrating the transfer device 20 that operates in the above-described modes A, B, C, D, E and F, respectively. At the separation position in each of the multiple modes A, B, C, D, E and F, the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T are moved downward in each of FIGS. 3A, 3B, 3C, 3D, 3E, and 3F, such that the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T are separated from the photoconductors 3K, 3Y, 3M, 3C, and 3T, respectively. Accordingly, the positions at which the intermediate transfer belt 2 is stretched by the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T change. The direction indicated by the double-headed arrow B is a direction in which the intermediate transfer belt 2 contacts with and separates from the multiple photoconductors 3K, 3Y, 3M, 3C, and 3T, and is also a direction in which the multiple primary transfer rollers 7K, 7Y, 7M, 7C, and 7T contact with and separate from the photoconductors 3K, 3Y, 3M, 3C, and 3T, respectively. The driven rollers 21A and 33A move downward in FIGS. 3A, 3B, 3C, 3D, 3E, and 3F in conjunction with movement of the primary transfer roller 7T of the most-downstream primary transfer section 203 in which the primary transfer roller 7T moves away from the photoconductor 3T. The driven rollers 21A and 33A move upward in FIGS. 3A, 3B, 3C, 3D, 3E, and 3F in conjunction with the movement of the primary transfer roller 7T in which the primary transfer roller 7T approaches the photoconductor 3T. The sensor 22 moves downward in FIGS. 3A, 3B, 3C, 3D, 3E, and 3F in accordance with the movement of the primary transfer roller 7T from the contact position or the large separation position to the small separation position. Movements of, for example, the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T in the above-described A, B, C, D, E and F are described below. Further, the position of the driven roller 33A indicated as the contact position in Table 1 is a first position, the position of the driven roller 33A indicated as the small separation position is a second position, and the position of the driven roller 33A indicated as the large separation position is a third position. Each of the primary transfer rollers 7K, 7Y, 7M, 7C, and 7T does not strictly move upward or downward in FIGS. 3A, 3B, 3C, 3D, 3E, and 3F.

(68) FIG. 4 is a diagram illustrating how the multiple modes A, B, C, D, E and F are switched, according to an embodiment of the present disclosure. An area surrounded by a solid line in FIG. 4 illustrates a case in which the modes A, B, C, D, and F are switched when the black (K) toner is arranged in the most-downstream primary transfer section 203. An area surrounded by a dotted line in FIG. 4 illustrates a case in which the modes A, B, D, and F are switched when the special color

toner is arranged in the most-downstream primary transfer section **203**. In other words, the mode C is a mode employed only when the black (K) toner is arranged in the most-downstream primary transfer section **203**, the mode E is a mode employed when the special color toner is arranged in the most-downstream primary transfer section **203** and switching between the mode C and the mode E is not performed.

(69) Switching between the modes A, B, C, D, F, and F as described above allows only the primary transfer sections to form the primary transfer nips needed for image formation. Accordingly, the primary transfer nips are not formed by the primary transfer sections that are not needed for image formation. Thus, excessive toner consumption can be prevented. For example, in the case in which a monochrome image is formed on a recording sheet, in the mode F, the black transfer nip NK is formed only in the most-upstream primary transfer section **201**. In particular, in the transfer device **20** according to the present embodiment in which the special color toner is transferred in the most-upstream primary transfer section **201** and the most-downstream primary transfer section **203**, the primary transfer roller **7K** of the most-upstream primary transfer section **201** and the primary transfer roller **7T** of the most-downstream primary transfer section **203** are contactable to and separable from the photoconductors **3K** and **3T**, respectively. By so doing, the primary transfer roller **7K** of the most-upstream primary transfer section **201** or the primary transfer roller **7T** of the most-downstream primary transfer section **203** can be separated from the photoconductor **3K**, or **3T**, respectively, as needed even when the special color toner is transferred either in the most-upstream primary transfer section **201** or the most-downstream primary transfer section **203**. Accordingly, excessive consumption of the special color toner can be prevented in any of the modes A, B, C, D, E, and F.

(70) When the multiple primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** are arranged at the respective contact positions and the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at the separation position, as indicated in the mode E, the primary transfer roller **7T** is arranged at the large separation position. Thus, the driven roller **33A** around which the intermediate transfer belt **2** is stretched is largely moved in the direction away from the photoconductor **3T**. As a result, the position at which the intermediate transfer belt **2** is stretched can be changed to a position away from the photoconductor **3T**. Such a configuration can prevent interference between the photoconductor **3T** and the intermediate transfer belt **2** and damage to the photoconductor **3T** and the intermediate transfer belt **2** due to the interference.

(71) In some switching operations among the switching operations between the modes A, B, C, D, E, and F, the order of components that contact with or separate from the intermediate transfer belt **2** is preset. Specifically, in the case in which the mode A is switched to the mode E, the primary transfer roller **7T** and the driven rollers **21A** and **33A** are moved first to the respective large separation positions. Then, the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** are moved to the respective contact positions. By contrast, in the case in which the mode E is switched to the mode A, the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** are moved first to the separation positions. Then, the primary transfer roller **7T** and the driven rollers **21A** and **33A** are moved to the respective small separation positions. In the above-described cases, the position of the primary transfer roller **7K** of the most-upstream primary transfer section **201** is switched between the separation position and the contact position at any suitable time. In the case in which the mode B is switched to the mode F, the primary transfer roller **7T** and the driven rollers **21A** and **33A** are moved first to the small separation position. Then, the primary transfer roller **7K** of the most-upstream primary transfer section **201** is moved to the contact position. By contrast, in the case in which the mode F is switched to the mode B, the primary transfer roller **7K** of the most-upstream primary transfer section **201** is moved first to the separation position. Then, the primary transfer roller **7T** and the driven rollers **21A** and **33A** are moved to the respective contact positions. In the case in which the

mode E is switched to the mode F, the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** are moved first to the respective separation positions. Then, the primary transfer roller **7T** and the driven rollers **21A** and **33A** are moved to the respective small separation positions. By contrast, in the case in which the mode F is switched to the mode E, the primary transfer roller **7T** and the driven rollers **21A** and **33A** are moved first to the large separation positions. Then, the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** are moved to the contact positions. As described above, the primary transfer roller **7T** and the driven rollers **21A** and **33A** that are moved to the separation position are moved first. Accordingly, damage to the intermediate transfer belt **2** and the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** due to the contact of the intermediate transfer belt **2** and the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** can be prevented.

(72) In a typical configuration in which multiple primary transferors of primary transfer sections other than a primary transfer section that transfers a toner image of a transparent color are moved to contact with and separated from corresponding one of multiple photoconductors, positions at which an intermediate transferor is stretched also change according to the arrangement of the above-described primary transferors. Accordingly, even if the primary transferor of the most-downstream primary transfer section is separated from the intermediate transferor, the intermediate transferor is not appropriately separated from the latent image bearer, which may cause damage to the intermediate transferor and the latent image bearer. However, according to the present embodiment, the intermediate transfer belt **2** serving as the intermediate transferor is properly separated from the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** serving as the latent image bearers. Thus, damage to the intermediate transfer belt **2** and the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** due to the contact of the intermediate transfer belt **2** and the photoconductors **3K**, **3C**, **3M**, **3Y**, and **3T** can be prevented.

(73) In the present embodiment, as described below, the primary transfer roller **7T** and the driven rollers **21A** and **33A** are simultaneously moved by a common moving mechanism. In some embodiments, when the primary transfer roller **7T** and the driven rollers **21A** and **33A** are moved by a different moving mechanism, the order in which the primary transfer roller **7T** and the driven rollers **21A** and **33A** are moved may be any desired order.

(74) A first contact-and-separation mechanism as a first movement mechanism that causes the primary transfer roller **7T** disposed in the most-downstream primary transfer section **203** to contact with and separate from the intermediate transfer belt **2** is described below. First, a motor that is a driving source of the first contact-and-separation mechanism and components surrounding the motor are described with reference to FIGS. **5** and **6**. FIG. **5** is a perspective view of a motor **23** and components surrounding the motor **23**, according to the present embodiment. FIG. **6** is a perspective view of the motor **23** and the components surrounding the motor **23** in which a bracket **29** covering a gear train is removed, according to the present embodiment.

(75) As illustrated in FIGS. **5** and **6**, the motor **23** that is a stepping motor is connected to a two-stage gear **24**. The two-stage gear **24** meshes with the motor **23** on teeth of one stage of the two-stage gear **24**, and the two-stage gear **24** rotates by the output of the motor **23**. Teeth of the other stage of the two-stage gear **24** mesh with teeth disposed on the shaft of a pulley **25** to transmit a driving force from the motor **23** to the pulley **25**. A toothed belt **26** is wound around the pulley **25** and a feeler-equipped pulley **27**. Teeth on an inner peripheral surface of the toothed belt **26** mesh with teeth on an outer peripheral surface of each of the pulley **25** and the feeler-equipped pulley **27**.

(76) The driving force of the motor **23** rotates a cam, to be described below, to cause the primary transfer roller **7T** (see FIGS. **28A**, **28B** and **28C**) to contact with or separate from the intermediate transfer belt **2**. The driving force of the motor **23** is also transmitted to the feeler-equipped pulley **27** via the two-stage gear **24**, the pulley **25**, and the toothed belt **26** to rotate the feeler-equipped pulley **27**.

(77) A photosensor **28** (see FIG. **7**) is disposed to face the feeler-equipped pulley **27**. Rotation of the feeler-equipped pulley **27** changes whether a feeler **27a** provided for the feeler-equipped pulley

27 is arranged at a position facing the photosensor 28. Thus, the feeler-equipped pulley 27 can change a condition in which the photosensor 28 detects. The photosensor 28 is attached to the bracket 29.

(78) FIG. 5 illustrates a case in which the primary transfer roller 7T is arranged at the small separation position, FIG. 7 illustrates a case in which the primary transfer roller 7T is arranged at the contact position, and FIG. 8 illustrates a case in which the primary transfer roller 7T is arranged at the large separation position. The motor 23 is driven by a predetermined number of pulses to rotate the feeler 27a counterclockwise to cause the primary transfer roller 7T to move from the large separation position at which the feeler 27a faces the photosensor 28 in FIG. 8. Subsequently, the motor 23 is stopped and held in a state in which the motor 23 can be driven to cause the primary transfer roller 7T to switch to the small separation position. Subsequently, the motor 23 is driven from the position in FIG. 8 by a predetermined number of pulses to rotate the feeler 27a clockwise. Subsequently, the motor 23 is stopped and held in the state in which the motor 23 can be driven to cause the primary transfer roller 7T to switch to the small separation position in FIG. 7. In other words, the position of the primary transfer roller 7T can be switched to the contact position and the small separation position via the large separation position. The positions of the primary transfer roller 7T, the driven rollers 21A and 33A, and the sensor 22 are switched between the small separation position, the contact position, and the large separation position by the driving force of the single motor 23.

(79) A first contact-and-separation mechanism 91 that causes the primary transfer roller 7T, the driven roller 21A, and the driven roller 33A to operate by the driving force of the motor 23 is described below with reference to FIG. 9. FIG. 9 is a cross-sectional view of the first contact-and-separation mechanism 91 viewed from a rear side of the image forming apparatus 1, which is an opposite side to the image forming apparatus 1 in, for example, FIG. 1.

(80) As illustrated in FIG. 9, the first contact-and-separation mechanism 91 includes a cam 31 to which the driving force of the above-described motor 23 is transmitted. The cam 31 includes a first cam 31A (see FIG. 12) and a second cam 31B and is rotatable about a rotation shaft 31a. The second cam 31B is a ball bearing having an outer ring. The second cam 31B is eccentric with respect to the rotation shaft 31a.

(81) The first cam 31A contacts a front slider 32 serving as a slider. As illustrated in FIG. 9, the front slider 32 is biased toward the left direction in FIG. 9 by springs. The driving force of the motor 23 causes the first cam 31A to rotate to change a surface of the first cam 31A that contacts the front slider 32. By so doing, the front slider 32 can move toward the right direction in FIG. 9 against the biasing force of the springs.

(82) The driven roller 33A, which is one of rollers around which the intermediate transfer belt 2 is stretched, is disposed at one end of the rotator 33. The rotator 33 is rotatable about a rotation fulcrum 33a. The rotator 33 includes a hole 33b at an end of the rotator 33 opposite to another end of the rotator 33 on which the driven roller 33A is disposed. An insertion portion 32a disposed on the front slider 32 is inserted into the hole 33b. The insertion portion 32a is formed by press-fitting a ball bearing into a shaft fixed to the front slider 32. Providing the ball bearings in the insertion portion 32a can reduce sliding resistance between the insertion portion 32a and the rotator 33. The primary transfer roller 7T is disposed at one end of a rotator 34. The rotator 34 is rotatable about a rotation fulcrum 34a. The rotator 34 includes a hole 34b at an end of the rotator 34 opposite to another end of the rotator 34 on which the primary transfer roller 7T is disposed. A pin 32b disposed on the front slider 32 is inserted into the hole 34b. A spring 35 is fixed to a housing of the image forming apparatus 1 and biases the rotator 34 in a direction in which the rotator 34 rotates clockwise in FIG. 9 about the rotation fulcrum 34a. The driven roller 33A is the first tension roller disposed downstream from the primary transfer roller 7T of the most-downstream primary transfer section 203 in the rotation direction of the intermediate transfer belt 2.

(83) When the front slider 32 moves in the left-right direction in FIG. 9, the insertion portion 32a

presses the rotator **33** to cause the rotator **33** to rotate about the rotation fulcrum **33a**. Accordingly, the position of the driven roller **33A** is changed. Further, when the front slider **32** moves in the right direction in FIG. **9**, the rotator **34** is pressed by the pin **32b** and rotates counterclockwise in FIG. **9** about the rotation fulcrum **34a** against the biasing force of the spring **35**. Alternatively, when the front slider **32** moves in the left direction in FIG. **9**, the rotator **34** rotates clockwise in FIG. **9** about the rotation fulcrum **34a** by the biasing force of the spring **35**. Thus, the primary transfer roller **7T** disposed on the rotator **34** contacts with and separates from the photoconductor **3T**.

(84) As illustrated in FIG. **9**, the driven roller **21A** around which the intermediate transfer belt **2** is stretched is disposed and driven by the rotation of the intermediate transfer belt **2**. The driven roller **21A** is disposed upstream from the primary transfer roller **7T** and downstream from the primary transfer roller **7C** immediately upstream from the primary transfer roller **7T** in the rotation direction of the intermediate transfer belt **2**. The driven roller **21A** is disposed at one end of the rotator **21**. The rotator **21** is rotatable about a rotation fulcrum **21a**. The rotator **21** receives a force from a spring **39** acting in a direction such that the rotator **21** rotates clockwise about the rotation fulcrum **21a**.

(85) In FIG. **9**, the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at the contact position. Under the above conditions, the front slider **32** is arranged at a leftmost position in FIG. **9** compared with the above-described other two positions at which the front slider **32** is arranged in FIG. **10** and FIG. **11**. When the first cam **31A** (see FIG. **12**) is rotated to a predetermined position to cause the primary transfer roller **7T** of the most-downstream primary transfer section **203** to be arranged at the small separation position, the front slider **32** moves to the right from the position of the front slider **32** in FIG. **9** to the position of the front slider **32** in FIG. **10**. Further, when the first cam **31A** is rotated to a predetermined position to cause the primary transfer roller **7T** of the most-downstream primary transfer section **203** to be arranged at the large separation position, the front slider **32** moves to the right from the position of the front slider **32** in FIGS. **9** and **10** to the position of the front slider **32** in FIG. **11**.

(86) For example, as illustrated in FIGS. **9**, **10**, and **11** in the order listed, when the front slider **32** moves in the right direction in FIG. **9**, the rotator **34** rotates counterclockwise about the rotation fulcrum **34a** against the biasing force of the spring **35**, and the primary transfer roller **7T** moves in the direction away from the photoconductor **3T**. When the primary transfer roller **7T** is arranged at the small separation position in FIG. **10** and at the large separation position in FIG. **11**, the primary transfer roller **7T** is separated from the photoconductor **3T**. When the front slider **32** moves in the right direction in FIG. **9**, the rotator **33** rotates counterclockwise about the rotation fulcrum **33a** and the driven roller **33A** moves in a direction away from the intermediate transfer belt **2**. The driven roller **33A** stretches the intermediate transfer belt **2** in all the configurations of FIGS. **9**, **10**, and **11**. However, the position at which the driven roller **33A** stretches the intermediate transfer belt **2** moves farther away from the photoconductor **3T**, which is the upper side of FIGS. **9**, **10**, and **11**, in the order of FIGS. **9**, **10**, and **11**. When the front slider **32** moves in the right direction in FIG. **9**, a pin **32c** (see FIG. **15**) disposed on the front slider **32** presses a side of the rotator **21** opposite to another side of the rotator **21** on which the driven roller **21A** is disposed. Accordingly, the rotator **21** rotates counterclockwise about the rotation fulcrum **21a** against the biasing force of the spring **39**. Thus, the driven roller **21A** moves away from the intermediate transfer belt **2** in FIGS. **10** and **11**.

(87) As described above, the position of the driven roller **33A** is changed in accordance with states in which the primary transfer roller **7T** is arranged: the contact position, the small separation position, or the large separation position. Accordingly, the position at which the driven roller **33A** stretches the intermediate transfer belt **2** can be changed depending on the state in which the primary transfer roller **7T** is arranged at the contact position, the small separation position, or the large separation position. As a result, the driven roller **33A** can stretch the intermediate transfer belt **2** at a favorable position, and the rotation speed of the intermediate transfer belt **2** can be accurately

detected by the sensor **22**. In particular, in the present embodiment, the driven roller **33A** is disposed downstream from the primary transfer roller **7T** of the most-downstream primary transfer section **203**. Accordingly, changing the position at which the driven roller **33A** stretches the intermediate transfer belt **2** in all of the above-described three states in which the primary transfer roller **7T**, the shape of the intermediate transfer belt **2** in which the intermediate transfer belt **2** is stretched in each of the three states can be appropriately changed. Accordingly, the sensor **22** can accurately detect the rotation speed of the intermediate transfer belt **2**. Furthermore, specifically in the above-described mode E in which the primary transfer roller **7T** is arranged at the large separation position, the rotator **33** is largely rotated counterclockwise in FIG. **11** to move the driven roller **33A** in the direction away from the photoconductor **3T**. By so doing, the position at which the intermediate transfer belt **2** is stretched by the driven roller **33A** can be shifted downward in FIG. **11**. In the mode E, the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** contact the intermediate transfer belt **2** to lift the intermediate transfer belt **2**. As a result, the intermediate transfer belt **2** is located at a position closer to the photoconductor **3T**. As described above, the position at which the intermediate transfer belt **2** is stretched by the driven roller **33A** is shifted downward in FIG. **11**. By so doing, the photoconductor **3T** (see FIG. **2**) and the intermediate transfer belt **2** can be prevented from being damaged due to interference between the photoconductor **3T** and the intermediate transfer belt **2**.

(88) A mechanism for moving the sensor **22** among mechanisms included in the first contact-and-separation mechanism **91** is described below.

(89) As illustrated in FIG. **9**, an outer circumferential surface of the second cam **31B** disposed in the cam **31** is held by a first arm **37** serving as a first link member or a second transmitter. The first arm **37** is rotatable about a rotation fulcrum **37a**. The rotation fulcrum **37a** is fixed to the front slider **32** via a ball bearing. As illustrated in FIG. **9**, the rotation of the second cam **31B** causes the first arm **37** to rotate about the rotation fulcrum **37a**. In addition, as the front slider **32** moves by rotation of the first cam **31A** (see FIG. **12**) disposed in the cam **31**, the first arm **37** moves in the left-right direction in FIG. **9**.

(90) FIG. **12** is a perspective view of the cam **31** according to the present embodiment. As illustrated in FIG. **12**, the cam **31** includes the first cam **31A** and the second cam **31B**. The cam **31** is rotatable about the rotation shaft **31a**. The first cam **31A** includes a small-diameter portion, a medium-diameter portion, and a large-diameter portion each having a different diameter by 120 degrees. As illustrated in FIG. **13**, the first cam **31A** is in contact with a cam follower **36** formed of a ball bearing. The cam follower **36** is a first transmitter provided for the first arm **37**. The rotation of the first cam **31A** changes a surface of the first cam **31A** that contacts the cam follower **36**. By so doing, the front slider **32** can be moved in the left-right direction in FIG. **9**. In addition, when the front slider **32** moves, the first arm **37** with the rotation fulcrum **37a** fixed to the front slider **32** moves in the left-right direction in FIG. **9** in conjunction with the movement of the front slider **32**.

(91) As illustrated in FIGS. **13** and **14**, the first arm **37** holds the second cam **31B** at two positions at which handle **37c1** and a handle **37c2** are disposed. The rotation of the second cam **31B** causes the first arm **37** to rotate about the rotation fulcrum **37a**.

(92) As illustrated in FIG. **13**, a thrust stopper **60** that serves as a restrictor and a slip-off stopper is attached to the first arm **37**. The thrust stopper **60** includes a contact portion **60a** and a restricting portion **60b** as slip-off stoppers. Bringing the contact portion **60a** into contact with the rotation fulcrum **37a** of the first arm **37** from above in FIG. **13** prevents the rotation fulcrum **37a** from coming off the front slider **32**. FIG. **14** is a side view of the first arm **37** in which the thrust stopper **60** is removed from the first arm **37**, according to the present embodiment. The thrust stopper **60** also contacts the rotation fulcrum **37a** from the lower side in FIG. **13** to prevent the rotation fulcrum **37a** from coming off in a downward direction in FIG. **13**. The restricting portion **60b** of the thrust stopper **60** is a surface of the thrust stopper **60** provided along the outer peripheral surface of the outer ring disposed on the second cam **31B** as the ball bearing. The restricting portion **60b**

regulates the position of the outer peripheral surface of the second cam **31B**. Accordingly, a direction in which the first arm **37** moves relative to the second cam **31B** can be restricted. In other words, the first arm **37** can be restricted from moving in a direction along the outer peripheral surface of the second cam **31B**, for example, in a direction in which the first arm **37** slides toward the second cam **31B**. Accordingly, the position of the first arm **37**, such as inclination of the first arm **37** with respect to the second cam **31B** can be prevented from being shifted, and wear of the handles **37c1** and **37c2** can be prevented.

(93) In the present embodiment, the contact portion **60a** that functions as the slip-off stopper to prevent the first arm **37** from coming off the front slider **32** and the regulating portion **60b** that regulates the direction in which the first arm **37** moves relative to the second cam **31B** are integrated with the thrust stopper **60**. Accordingly, the number of components of the transfer device **20** can be reduced. However, the contact portion **60a** and the regulating portion **60b** may be disposed as separate components.

(94) FIG. **15** is a perspective view of the first arm **37**, a second arm **38**, and components around the first arm **37** and the second arm **38** viewed from a front side of the image forming apparatus **1**, according to the present embodiment. FIG. **16** is a perspective view of the first arm **37**, the second arm **38**, and components around the first arm **37** and the second arm **38** viewed from a rear side of the image forming apparatus **1**, according to the present embodiment.

(95) As illustrated in FIG. **15**, the second arm **38** that serves as a second link member includes an elongated hole **38a** and an elongated hole **38b** at each of both ends of the second arm **38**. An end **37b** of the first arm **37** is inserted into the elongated hole **38a** of the second arm **38**. As illustrated in FIG. **16**, the end **37b** of the first arm **37** includes a bearing **40**. The bearing **40** is disposed to be movable in the elongated hole **38a** in a longitudinal direction of the elongated hole **38a**. The bearing **40** serves as an insertion portion through which the elongated hole **38a** is inserted.

(96) The bearing **40** includes a parallel pin **40a** that serves as a slip-off stopper in a rear portion of the bearing **40**. The length of the parallel pin **40a** is set to be shorter than the length of the elongated hole **38a** in the longitudinal direction of the elongated hole **38a**. Arranging the parallel pin **40a** substantially parallel to the longitudinal direction of the elongated hole **38a** allows the bearing **40** to be inserted into the elongated hole **38a**. As described above, in the three states in which the primary transfer roller **7T** is arranged at the contact position, the small separation position, and the large separation position, the parallel pin **40a** does not rotate to a position at which the parallel pin **40a** is parallel to the longitudinal direction of the elongated hole **38a**. Accordingly, the parallel pin **40a** functions as the slip-off stopper to prevent the bearing **40** from coming off the elongated hole **38a**.

(97) As illustrated in FIG. **15**, a bearing **41** is inserted into the elongated hole **38b**. The bearing **41** is fixed to a first sensor bracket **43** as a holder by a step screw **42**. The bearing **41** is movable in the elongated hole **38b**. The bearing **41** serves as an insertion portion through which the elongated hole **38b** is inserted.

(98) Rotation of the cam **31** causes the front slider **32** to be moved from the position of the front slider **32** in FIG. **9** or FIG. **10** to the right side of FIG. **10** to cause the primary transfer roller **7T** of the most-downstream primary transfer section **203** to move to the large separation position. By so doing, the second cam **31B** rotates to cause the first arm **37** to rotate clockwise about the rotation fulcrum **37a**. Accordingly, the end **37b** of the first arm **37** moves downward in FIG. **9** or **10**. Accordingly, as illustrated in FIG. **11**, the end **37b** moves to an end of the elongated hole **38a** in the longitudinal direction and contacts a wall surface forming the elongated hole **38a** to pull the second arm **38** in a lower left direction in FIG. **11**. Accordingly, the bearing **41** move to an end of the elongated hole **38b** in the longitudinal direction and contact a wall surface forming the elongated hole **38b**. Then, the second arm **38** pulls the first sensor bracket **43** in the lower left direction in FIG. **11**.

(99) FIG. **17** is a diagram illustrating a configuration or structure around the first sensor bracket **43**

and a sensor **22** and is a diagram in which the rotator **21** is removed from FIG. **9**, according to the present embodiment. In FIG. **17**, the sensor **22** and a second sensor bracket **44** are illustrated in a simplified manner for the sake of convenience.

(100) As illustrated in FIG. **17**, the first sensor bracket **43** is rotatable about the rotation fulcrum **43a**. The first sensor bracket **43** receives a force from the spring **45** fixed to the housing of the image forming apparatus **1** in a direction in which the first sensor bracket **43** rotates counterclockwise in FIG. **17** about the rotation fulcrum **43a**. A restrictor **63** is fixed to the first sensor bracket **43**. The pin **32d** of the front slider **32** is inserted into a hole **63a** of the restrictor **63**. When the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at the contact position in FIG. **9** and at the small separation state in FIG. **10**, the pin **32d** contacts a wall surface forming walls of the hole **63a**. By so doing, the front slider **32** applies a force to the first sensor bracket **43** such that the first sensor bracket **43** rotates clockwise about the rotation fulcrum **43a** in FIG. **17**.

(101) The second sensor bracket **44** is fixed to the first sensor bracket **43** via a stud **43b** disposed on the first sensor bracket **43**. The second sensor bracket **44** holds the sensor **22**. The second sensor bracket **44** includes a hook **44a** to which one end of a spring **62** (see FIG. **9**) is attached, a first contact portion **44b**, and a second contact portion **44c**.

(102) When the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at the contact position in FIG. **9**, the second sensor bracket **44** is biased by the spring **62** to move in a direction in which the second sensor bracket **44** rotates clockwise about the rotation fulcrum **43a** and is positioned at a position at which the first contact portion **44b** contacts a stud **64** disposed on the housing of the image forming apparatus **1**.

(103) On the other hand, in the state in which the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at the small separation position in FIG. **10**, the pin **32d** disposed on the front slider **32** moves to the right side of FIG. **9**. By so doing, the first sensor bracket **43**, the second sensor bracket **44**, and the sensor **22** receive a force to rotate counterclockwise about the rotation fulcrum **43a** due to their own weight and the biasing force of the spring **45**. A pin **43c** disposed on the first sensor bracket **43** illustrated in FIG. **18** presses a bent portion **44d** of the second sensor bracket **44**, and the second sensor bracket **44** receives a force such that the second sensor bracket **44** rotates counterclockwise about the rotation fulcrum **43a** in FIG. **10**. Accordingly, the first sensor bracket **43**, the second sensor bracket **44**, and the sensor **22** rotate counterclockwise in FIG. **10** and move downward in FIG. **10**, which is a direction away from the photoconductor **3** as compared with FIG. **9**. Note that FIG. **18** is a perspective view of the first sensor bracket **43** and the second sensor bracket **44** viewed from the back side of the sheet surface of FIG. **10**.

(104) When the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at the large separation position in FIG. **11**, the pin **32d** is further moved rightward to release a force of the pin **32d** pressing the restrictor **63** leftward in FIG. **17** as illustrated in FIG. **17**. At the same time, the second arm **38** pulls the first sensor bracket **43** in a lower left direction in FIG. **17** as described above to rotate the first sensor bracket **43** clockwise about the rotation fulcrum **43a** in FIG. **17**. Accordingly, the second sensor bracket **44** fixed to the first sensor bracket **43** via the stud **43b** moves upward in FIG. **17**, and the sensor **22** also moves upward in FIG. **17**. At this time, as illustrated in FIG. **19**, the second sensor bracket **44** is positioned at a position at which the second contact portion **44c** of the second sensor bracket **44** contacts a positioning portion **21b** of the rotator **21**. In other words, the upward movement of the second sensor bracket **44** and the sensor **22** in FIG. **17** is restricted, and the sensor **22** is positioned. In the present embodiment, the above-described position at which the sensor **22** is positioned is a position lower than a position of the sensor **22** when the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at the contact position in FIG. **9** and upper than the position of the sensor **22** when the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at

the small separation position in FIG. 10.

(105) As described above, the driving force of the cam **31** is transmitted to the first sensor bracket **43** via the link members such as the first arm **37** and the second arm **38** to rotate the first sensor bracket **43**. By so doing, the first sensor bracket **43** can be rotated in a desired direction.

(106) In particular, in the present embodiment, the rotational force of the first arm **37** is transmitted to the second arm **38** only when a predetermined condition is satisfied. Accordingly, the driving force by the rotation of the cam **31** to the sensor **22** can be transmitted only when a specific positional change is performed. To be more specific, the second arm **38** is connected to the first arm **37** and the first sensor bracket **43** via the elongated holes **38a** and **38b**, respectively, disposed in the second arm **38**. Accordingly, the second arm **38** can be retracted to move the sensor **22** downward, for example, in FIG. 11 only when the primary transfer roller 7T of the most-downstream primary transfer section **203** is arranged at the large separation position. In other words, compared with the primary transfer roller 7T and the driven rollers **21A** and **33A** that move in a constant direction with the movement of the front slider **32**, the sensor **22** moves upward in FIG. 10 when the primary transfer roller 7T of the most-downstream primary transfer section **203** moves from the contact position to the small separation position and moves downward. Alternatively, the sensor **22** moves downward, for example, in FIG. 11 when the primary transfer roller 7T of the most-downstream primary transfer section **203** moves from the contact position to the large separation position or from the small separation position to the large separation position. Thus, the sensor **22** moves in a direction opposite to the direction in which the front slider **32** moves. When the primary transfer roller 7T of the most-downstream primary transfer section **203** is arranged at the small separation position, the primary transfer roller 7K of the most-upstream primary transfer section **201** and the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section **202** separate from the photoconductors **3K**, **3Y**, **3M**, and **3C**, respectively. As a result, the position at which the intermediate transfer belt **2** is stretched moves downward in FIG. 10. On the other hand, when the primary transfer roller 7T of the most-downstream primary transfer section **203** is arranged at the large separation position, the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section **202** contact the photoconductors **3Y**, **3M**, and **3C**, respectively, via the intermediate transfer belt **2**. As a result, the position at which the intermediate transfer belt **2** is stretched is pushed upward in FIG. 11. Accordingly, changing the position of the sensor **22** as described above allows the sensor **22** to be positioned at a favorable position corresponding to the position at which the intermediate transfer belt **2** is stretched. Accordingly, in each of the modes A, B, C, D, E, and F, the detection accuracy of the sensor **22** with respect to the intermediate transfer belt **2** can be enhanced, and the traveling speed of the intermediate transfer belt **2** can be controlled with high accuracy. Further, the first contact-and-separation mechanism **91** can perform the operation of the sensor **22** and the operations of the primary transfer roller 7T and the driven rollers **21A** and **33A** by the driving force of the motor **23** as a single driving source. Accordingly, energy saving and a reduction in the number of components of the transfer device **20** can be achieved.

(107) However, the number of link members coupled to the first sensor bracket **43** holding the sensor **22** is not limited to two as in embodiments of the present disclosure. The number of the link members may be three or greater than or one. Further, the combination of the elongated hole and the insertion member such as a pin inserted into the elongated hole may be reversed. It is not necessarily need to operate all of the sensor **22**, the primary transfer roller 7T, and the driven rollers **21A** and **33A** by the driving force of the motor **23**.

(108) As described above, in the present embodiment, the rotation of the first cam **31A** illustrated in FIG. 12 causes the front slider **32** to move in the left-right direction in FIG. 9. By so doing, the primary transfer roller 7T and the driven rollers **21A** and **33A** can be moved. Further, the sensor **22** can be moved by the rotation of the first cam **31A** and the second cam **31B**. To be specific, when the primary transfer roller 7T of the most-downstream primary transfer section **203** is arranged at the contact position in FIG. 9 and the small separation position in FIG. 10, the pin **32d** (see FIG.

16) is moved by the rotation of the first cam 31A to press the first sensor bracket 43. Accordingly, a force that causes the first sensor bracket 43 to rotate clockwise is applied to the first sensor bracket 43. As a result, the position of the sensor 22 can be changed. When the primary transfer roller 7T of the most-downstream primary transfer section 203 is arranged at the large separation position, the rotation of the second cam 31B causes the first sensor bracket 43 to be pulled by the second arm 38. Accordingly, the position of the sensor 22 can be changed.

(109) A second contact-and-separation mechanism 92 as a second movement mechanism and a third contact-and-separation mechanism 93 as a third movement mechanism are described below with reference to FIG. 20. The second contact-and-separation mechanism 92 causes the primary transfer rollers 7C, 7M and 7Y disposed in the central primary transfer section 202 to contact with and separate from the intermediate transfer belt 2. The third contact-and-separation mechanism 93 causes the primary transfer roller 7K disposed in the most-upstream primary transfer section 201 to contact with and separate from the intermediate transfer belt 2.

(110) As illustrated in FIG. 20, the second contact-and-separation mechanism 92 includes rotators 46, 47, 48, a cam 51, and a cam follower 52. The third contact-and-separation mechanism 93 includes a rotator 49, a cam 53, and a cam follower 54. The second contact-and-separation mechanism 92 includes a motor as a driving source to rotate the cam 51, and the third contact-and-separation mechanism 93 includes a motor as a driving source to rotate the cam 53.

(111) The rotators 46, 47, 48, and 49 are rotatable about the rotation fulcrums 46a, 47a, 48a, and 49a, respectively. The primary transfer roller 7C is disposed at one end of the rotator 46. The primary transfer roller 7M is disposed at one end of the rotator 47. The primary transfer roller 7Y is disposed at one end of the rotator 48. The primary transfer roller 7K is disposed at one end of the rotator 49. The rotators 46, 47, 48, and 49 are biased by springs to be rotated in a direction in FIG. 20 and cause the primary transfer rollers 7C, 7M, 7Y, and 7K, respectively, to contact the photoconductors 3C, 3M, 3Y, and 3K, respectively, via the intermediate transfer belt 2.

(112) The cam follower 52 rotates by the rotation of the cam 51 to move a front slider 50 of the most-upstream primary transfer section 201 in the right direction in FIG. 20. Accordingly, one end of each of the rotators 46, 47, and 48 opposite to another end at which the corresponding one of the primary transfer rollers 7C, 7M, and 7Y is disposed is pressed. Accordingly, the rotators 46, 47, and 48 rotate counterclockwise in FIG. 20 against the biasing force of the springs. Accordingly, the primary transfer rollers 7C, 7M, and 7Y move away from the intermediate transfer belt 2. Further, the rotation of the cam 53 causes the cam follower 54 to rotate and one end of the rotator 49 opposite to another end of the rotator 49 at which the primary transfer roller 7K is disposed is pressed. Accordingly, the rotator 49 rotates counterclockwise in FIG. 20 against the biasing force of the spring, and the primary transfer roller 7K moves away from the intermediate transfer belt 2. As described above, the primary transfer roller 7K of the most-upstream primary transfer section 201 and the primary transfer rollers 7C, 7M, and 7Y of the central primary transfer section 202 independently contact with and separate from the intermediate transfer belt 2.

(113) Toner supply devices that supply toner to the developing devices 6Y, 6M, 6C, 6K, and 6T are described below with reference to FIGS. 21, 22, and 23.

(114) As illustrated in FIG. 21, the image forming apparatus 1 includes a bottle container 101 in an upper portion of the housing of the image forming apparatus 1. Toner bottles 102Y, 102M, 102C, 102K, and 102T that contain yellow (Y) toner, magenta (M) toner, cyan (C) toner, black (K) toner, and special color toner, respectively, to be supplied are attached to the bottle container 101. Bottle drivers 103Y, 103M, 103C, 103K, and 103T (see FIG. 23) of the toner supply device 150 are fixed to the bottle container 101. The bottle drivers 103Y, 103M, 103C, 103K, and 103T detachably holds the toner bottles 102Y, 102M, 102C, 102K, and 102T, respectively.

(115) FIG. 23 is a schematic diagram illustrating a configuration of a toner supply device 150 according to the present embodiment. In FIG. 23, a toner bottle 102, a toner supply device 150, a developing device 6, and a photoconductor 3 for one of the colors T, Y, M, C, or K are illustrated.

In FIG. 23 and in the description below, suffixes T, Y, M, C, and K attached to the reference signs of the toner bottle **102**, the toner supply device **150**, the developing device **6**, and the photoconductor **3** are omitted for the sake of convenience.

(116) The toner supply device **150** includes a bottle driver **103**, a pre-supply reservoir **104**, a toner supply unit **105**, a suction pump **106**, and a transfer tube **107**. The pre-supply reservoir **104** is disposed directly above the developing device **6**. One end of transfer tube **107** is connected to the bottle driver **103** and the other end of the transfer tube **107** is connected to the suction pump **106** to form a toner conveyance path to transfer toner from the bottle driver **103** to the pre-supply reservoir **104**. In the present embodiment, the transfer tube **107** is a flexible tube.

(117) The bottle driver **103** drives the toner bottle **102** to rotate. Accordingly, toner contained in the toner bottle **102** is transferred from a head opening of the toner bottle **102** to the bottle driver **103**. Suction operation of the suction pump **106** causes the toner in the bottle driver **103** to be transferred to the suction pump **106** via the transfer tube **107**. At the same time, the toner sucked from the bottle driver **103** is dropped into the pre-supply reservoir **104** from a discharge port of the suction pump **106**.

(118) Rotation of the toner supply unit **105** causes the toner stored in the pre-supply reservoir **104** to be supplied to the developing device **6** via a toner supply path **108**. As described above, in the present embodiment, the toner transferred from the bottle driver **103** to the vicinity of the developing device **6** by the suction pump **106** is temporarily stored in the pre-supply reservoir **104**.

(119) Note that, for example, in the case in which a white toner is employed as a special color to form a white background in an image, a white toner layer is formed at a lowermost layer of the image. For this reason, the most-downstream primary transfer section **203** is arranged most downstream among the most-upstream primary transfer section **201**, the central primary transfer section **202**, and the most-downstream primary transfer section **203**. Alternatively, when a transparent toner image is transferred to apply glossiness to an image, the transparent toner image is formed on the surface of the image. For this reason, in this case, the most-downstream primary transfer section **203** is arranged most upstream among the most-upstream primary transfer section **201**, the central primary transfer section **202**, and the most-downstream primary transfer section **203**.

(120) As described above, in order to change the order in which the toner of the special color is primarily transferred in accordance with the type of the special color to be employed, in the present embodiment, the toner supplied to the most-upstream primary transfer section **201** and the most-downstream primary transfer section **203** can be changed as illustrated in FIGS. 21 and 22. More specifically, in FIG. 21, the toner bottle **102T** that contains a special color toner is connected to a pre-supply reservoir **104T** disposed most downstream in the toner conveyance path on the right side of FIG. 21. The toner bottle **102K** that contains black (K) toner is connected to a pre-supply reservoir **104K** disposed most upstream in the toner conveyance path in FIG. 21. In FIG. 22, the toner bottle **102K** that contains the black (K) toner is connected to the pre-supply reservoir **104K** disposed most downstream in the toner conveyance path on the right side of the FIG. 22. The toner bottle **102T** that contains the special color toner is connected to the pre-supply reservoir **104T** disposed most upstream in the toner conveyance path. In addition, in FIGS. 21 and 22, the arrangement of the toner bottles **102K** and **102T** and the transfer tubes **107K** and **107T** connected to the toner bottles **102K** and **102T**, respectively, are not changed, and destinations to which the transfer tubes **107K** and **107T** are connected are changed. In other words, in FIG. 21, the transfer tube **107K** is extended and connected to the suction pump **106T** disposed most upstream in the toner conveyance path. On the other hand, in FIG. 22, the transfer tube **107T** is connected to the suction pump **106T** disposed most downstream in the toner conveyance path. In contrast to the transfer tube **107T**, the transfer tube **107T** is significantly stretched toward upstream in the toner conveyance path in FIG. 22. Accordingly, positions at which the black (K) toner and the special color toner are primarily transferred can be changed only by changing the positions of the pre-

supply reservoirs **104K**, **104Y**, **104M**, **104C**, and **104T** and the image forming devices **10K**, **10Y**, **10M**, **10C**, and **10T** without replacing the toner bottles **102K**, **102Y**, **102M**, **102C**, and **102T** or the bottle drivers **103Y**, **103M**, **103C**, **103K**, and **103T**. As a result, the labor for replacing the toner bottles **102K**, **102Y**, **102M**, **102C**, and **102T** or the bottle drivers **103Y**, **103M**, **103C**, **103K**, and **103T** can be reduced. The actual length of the transfer tube **107T** is longer than the length of the transfer tube **107T** illustrated in FIG. 21 and the actual length of the transfer tube **107K** is longer than the length of the transfer tube **107K** illustrated in FIG. 22. Accordingly, a space in which extra tubes of the transfer tube **107T** and the transfer tube **107K** can be accommodated is disposed in the image forming apparatus **1**.

(121) The operation of changing the colors of toner transferred by the most-upstream primary transfer section **201**, the central primary transfer section **202**, and the most-downstream primary transfer section **203** is described below with reference to the flowchart of FIG. 24.

(122) As illustrated in FIG. 24, first, setting of the arrangement of colors of toner is changed (step S1). Specifically, the black (K) toner is arranged in the most-upstream primary transfer section **201** and the special color toner is arranged in the most-downstream primary transfer section **203**.

Alternatively, the special color toner is arranged in the most-upstream primary transfer section **201** and the black (K) toner is arranged in the most-downstream primary transfer section **203**. Then, a controller **300** (see FIG. 25) of the image forming apparatus **1** determines whether the colors of toner are correctly arranged. When the colors of toner are not correctly arranged, the controller **300** displays a message prompting to replace the black (K) toner with the special color toner on an operation display unit (steps S2 and S3). Then, the power supply of the image forming apparatus **1** is turned off and the pre-supply reservoir **104K** is replaced with the pre-supply reservoir **104T** and the image forming device **10K** is replaced with image forming device **10T**. Subsequently, the power supply of the image forming apparatus **1** is turned on again (steps S4, S5, and S6). Then, the controller **300** of the image forming apparatus **1** determines again whether the pre-supply reservoir **104K** has been replaced with the pre-supply reservoir **104T** and the image forming device **10K** have been replaced with image forming device **10T** (step S7) correctly. If the replacement has not been correctly performed, the message to replace the black (K) toner with the special color toner is displayed again on the operation display unit (step S8).

(123) In steps S2, S3, S4, S5, S6, and S7, the controller **300** determines whether the black (K) toner and the special color toner are correctly arranged. At the same time, the controller **300** also determines whether a correct color such as the transparent color or the white color is set as the special color.

(124) Before the setting of the image forming apparatus **1** is changed, the power supply of the image forming apparatus **1** may be turned off as in step S4 and the arrangement of the toner colors may be changed as in step S5.

(125) As illustrated in FIG. 25, the controller **300** provided for the image forming apparatus **1** determines whether the black (K) toner and the special color toner are correctly arranged in steps S1, S2, S3, S4, S5, and S6. The controller **300** includes a determination circuit **301** that determines whether the pre-supply reservoirs **104K**, **104Y**, **104M**, **104C**, and **104T** and the image forming devices **10K**, **10C**, **10M**, **10Y**, and **10T** are properly arranged.

(126) The determination circuit **301** includes a first connector **302**, a second connector **303**, a third connector **304**, and a fourth connector **305**. The first connector **302** is connected to the pre-supply reservoir **104K** disposed most upstream in the toner conveyance path. The second connector **303** is connected to the pre-supply reservoir **104T** disposed most downstream in the toner conveyance path. The third connector **304** is connected to the image forming device **10K** disposed most upstream in the toner conveyance path. The fourth connector **305** is connected to the image forming device **10T** disposed most downstream in the toner conveyance path. The pre-supply reservoir **104K** includes a circuit board **104K1** connected to the first connector **302**, and the pre-supply reservoir **104T** includes a circuit board **104T1** connected to the second connector **303**. A

circuit board **10K1** connected to the third connector **304** is disposed in, for example, a developer container of the developing device **6K** of the image forming device **10K**. A circuit board **10T1** connected to the fourth connector **305** is disposed in, for example, a developer container of the developing device **6T** of the image forming device **10T**.

(127) The first connector **302**, the second connector **303**, the third connector **304**, and the fourth connector **305** each includes multiple switches. The determination circuit **301** can determine whether the black (K) toner or the special color toner is arranged and the color of the special color toner if the special color toner is arranged, based on a combination of on and off of the switches when the circuit board **104K1**, the circuit board **104T1**, the first connector **302**, the second connector **303**, the third connector **304**, the circuit board **10K1**, and the circuit board **10T1** is each connected to the first connector **302**, the second connector **303**, the third connector **304**, and the fourth connector **305**, respectively. In a case in which the controller **300** determines only whether the black (K) toner or the special color toner is arranged without determining the color of the special color toner, the controller **300** may perform the determination based on whether the feeler **27a** and the photosensor **28** are turned on or off.

(128) The controller **300** receives a detection result of the sensor **22**. The controller **300** changes the rotation speed of the intermediate transfer belt **2** based on the detection result.

(129) The transfer device **20** according to a modification of the above-described embodiments is described below with reference to FIGS. **26** and **27**. FIG. **27A** is a diagram illustrating the second contact-and-separation mechanism **92** in which the primary transfer rollers **7C**, **7M**, and **7Y** are arranged at the contact positions to contact the intermediate transfer belt **2**, according to the modification. FIG. **27B** is a diagram illustrating the second contact-and-separation mechanism **92** in which the primary transfer rollers **7C**, **7M**, and **7Y** are arranged at the respective separation positions separated from the intermediate transfer belt **2**, according to the modification.

(130) As illustrated in FIG. **26**, in the present modification, a driven roller **55A** that stretches the intermediate transfer belt **2** is disposed between the primary transfer roller **7T** disposed in the most-downstream primary transfer section **203** and the primary transfer roller **7C** upstream from the primary transfer roller **7T** in the rotation direction of the intermediate transfer belt **2**. The driven roller **55A** is disposed upstream from the sensor **22** in the rotation direction of the intermediate transfer belt **2**. As illustrated in FIG. **27A**, the driven roller **55A** is disposed at one end of the rotator **55**. The rotator **55** is rotatable about a rotation fulcrum **55a** disposed at an end of the rotator **55** opposite to another end of the rotator **55** at which the driven roller **55A** is disposed.

(131) The second contact-and-separation mechanism **92** includes the cam **51**. The rotation fulcrum **55a** is fixed to the front slider **50** that causes the primary transfer rollers **7C**, **7M**, and **7Y** of the central primary transfer section **202** to contact with or separate from the intermediate transfer belt **2**. When the cam **51** rotates to move the front slider **50** to the right in FIG. **27A**, the rotator **55** is rotated clockwise about the rotation fulcrum **55a** as illustrated in FIG. **27B**.

(132) In the above-described embodiment, the driven roller **55A** contacts the intermediate transfer belt **2** when the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** are positioned at the respective contact positions to stretch the intermediate transfer belt **2**. In the above-described mode E in which the primary transfer roller **7T** of the most-downstream primary transfer section **203** is arranged at the large separation position and the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** are arranged at the contact position, the primary transfer roller **7T** of the most-downstream primary transfer section **203** is separated from the photoconductor **3T**. For this reason, nip pressure of the multiple transfer nips of the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** is likely to be small. In the present modification, the driven roller **55A** disposed between the primary transfer roller **7T** of the most-downstream primary transfer section **203** and the primary transfer roller **7C** disposed immediately upstream from the primary transfer roller **7T** contacts the intermediate transfer belt **2** when the primary transfer rollers **7Y**, **7M**, and **7C** of the central primary transfer section **202** are

arranged at the respective contact positions. By so doing, transfer pressure of the transfer nips of the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 can be prevented from being decreased.

(133) In addition, the sensor 22 is disposed between the driven roller 55A and the primary transfer roller 7T. By so doing, the rotation speed of the intermediate transfer belt 2 can be detected in a state in which there is no influence of the vibration of the driven roller 55A to the intermediate transfer belt 2. Thus, accuracy of the rotation speed of the intermediate transfer belt 2 in the most-downstream primary transfer section 203 can be particularly enhanced.

(134) Another embodiment of the present disclosure is described below with reference to FIGS. 28A, 28B, and 28C. In the present embodiment, the driven roller 56A, which is disposed between the primary transfer roller 7T of the most-downstream primary transfer section 203 and the primary transfer roller 7C immediately upstream from the primary transfer roller 7T, is moved by the first contact-and-separation mechanism 91 that causes the primary transfer roller 7T of the most-downstream primary transfer section 203 to contact with or separate from the intermediate transfer belt 2. FIG. 28A is a diagram illustrating the first contact-and-separation mechanism 91 in a case in which the primary transfer roller 7T is arranged at the contact position, according to the present embodiment. FIG. 28B is a diagram illustrating the first contact-and-separation mechanism 91 in a case in which the primary transfer roller 7T is arranged at the small separation position, according to the present embodiment. FIG. 28C is a diagram illustrating the first contact-and-separation mechanism 91 in a case in which the primary transfer roller 7T is arranged at the large separation position, according to the present embodiment.

(135) As illustrated in FIG. 28A, a rotator 56 is rotatable about a rotation fulcrum 56a disposed at an end of the rotator 56 opposite to another end of the rotator 56 at which the driven roller 56A is disposed. The rotation fulcrum 56a is fixed to the front slider 32 that causes the primary transfer roller 7T of the most-downstream primary transfer section 203 to contact with or separate from the intermediate transfer belt 2. A mechanism that causes the sensor 22 to contact with or separate from the intermediate transfer belt 2 is similar to the mechanism employed in the above-described embodiment.

(136) The rotator 56 includes a hole 56b. A pin 32e of the front slider 32 is inserted into the hole 56b. The hole 56b has the same height at both ends of the hole 56b in the horizontal direction in FIGS. 28A, 28B, and 28C, i.e., a direction in which the front slider 32 moves. The height is a height of the hole 56b in the vertical direction in FIGS. 28A, 28B, and 28C and height of the hole 56b in a direction in which the hole 56b contacts with or moves away from the intermediate transfer belt 2. In addition, the hole 56b has a shape such that the hole 56b includes a convex portion 56b1 protruding toward the intermediate transfer belt 2 at the center of the hole 56b in the horizontal direction in FIGS. 28A, 28B, and 28C, i.e., the direction in which the front slider 32 moves. Due to the shape of the hole 56b described above, the driven roller 56A can be separated from the intermediate transfer belt 2 only when the primary transfer roller 7T of the most-downstream primary transfer section 203 is arranged at the small separation position. The driven roller 56A can contact the intermediate transfer belt 2 when the primary transfer roller 7T of the most-downstream primary transfer section 203 is arranged at the large separation position. In other words, when the primary transfer roller 7T of the most-downstream primary transfer section 203 in FIG. 28B is arranged at the small separation position, the pin 32e of the front slider 32 is accommodated in the convex portion 56b1 of the hole 56b. Accordingly, the rotator 56 rotates clockwise in FIG. 28B. As a result, the driven roller 56A moves away from the intermediate transfer belt 2. On the other hand, when the primary transfer roller 7T of the most-downstream primary transfer section 203 in FIG. 28B is arranged at the large separation position, the pin 32e moves toward a right end of the hole 56b. Accordingly, the rotator 56 rotates counterclockwise, and the driven roller 56A contacts the intermediate transfer belt 2.

(137) Also in the present embodiment, the driven roller 56A can contact the intermediate transfer

belt 2 when the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 are arranged at the respective contact positions and the primary transfer roller 7T of the most-downstream primary transfer section 203 is arranged at the large separation position. As a result, the transfer pressure of the transfer nips of the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 can be prevented from being decreased. Further, due to the shape of the hole 56d described above, the driven roller 56A can be separated from the intermediate transfer belt 2 only when the primary transfer roller 7T of the most-downstream primary transfer section 203 is arranged at the small separation position.

(138) The primary transfer roller 7T of the most-downstream primary transfer section 203 may be switched only between the two positions of the contact position and the large separation position described in the above-described embodiments. However, in this case, the primary transfer roller 7T of the most-downstream primary transfer section 203 is arranged at the large separation position and the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 are arranged at the respective separation positions. In such a combination, remaining belt length of the intermediate transfer belt 2 needs to be adjusted.

(139) FIG. 29 is a diagram illustrating the transfer device 20 operated in the above-described mode D in which the primary transfer roller 7T of the most-downstream primary transfer section 203 and the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 are arranged at the respective contact positions, according to the present embodiment.

(140) In FIG. 29, the primary transfer roller 7K of the most-upstream primary transfer section 201, the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202, the primary transfer roller 7T of the most-downstream primary transfer section 203, and driven roller 33A contact the intermediate transfer belt 2 from below in FIG. 29, to stretch the intermediate transfer belt 2. From the above-described state, the primary transfer roller 7T of the most-downstream primary transfer section 203 is moved to the large separation position and the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 are moved to the separation position. In other words, the multiple primary transfer rollers 7Y, 7M, 7C, and 7T and the driven roller 33A are moved downward in FIG. 29 to separate the intermediate transfer belt 2 from the photoconductors 3Y, 3M, 3C, and 3T. Accordingly, a difference between the circumferential length of the intermediate transfer belt 2 is larger than the circumferential length of the intermediate transfer belt 2 in FIG. 29. For this reason, the intermediate transfer belt 2 has a surplus length.

(141) For this reason, in the present embodiment, a position varying mechanism that allows the position of a tension roller 65 to be variable is provided for the intermediate transfer belt 2. The tension roller 65 applies tension to the intermediate transfer belt 2 from the outer circumferential surface of the intermediate transfer belt 2. This position varying mechanism is described with reference to FIGS. 30A and 30B below. FIG. 30A is a diagram illustrating the tension roller 65 attached to the rotation mechanism 66 in the mode D in which the primary transfer roller 7T of the most-downstream primary transfer section 203 and the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 are arranged at the contact position, according to the present embodiment. FIG. 30B is a diagram illustrating the tension roller 65 attached to the rotation mechanism 66 in which the primary transfer roller 7T of the most-downstream primary transfer section 203 is arranged at the large separation position and the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 are arranged at the separation position, according to the present embodiment. The other mechanisms of the present embodiment are similar to the mechanisms of the above-described embodiments.

(142) As illustrated in FIG. 30A, the tension roller 65 is attached to one end of the rotation mechanism 66. The rotation mechanism 66 is rotatable around a rotation fulcrum 66a. One end of a spring 67 is fixed to the other end of the rotation mechanism 66. The other end of the spring 67 is fixed to a housing of the image forming apparatus 1 by a stud 68. A force is applied to the rotation mechanism 66 by the spring 67 to cause the rotation mechanism 66 to rotate counterclockwise in

FIG. 30A about the rotation fulcrum 2a.

(143) For example, when the primary transfer rollers 7Y, 7M, and 7C of the central primary transfer section 202 and the primary transfer roller 7T of the most-downstream primary transfer section 203 in FIG. 29 are moved from the contact position to the separation position or the large separation position, the position at which the intermediate transfer belt 2 is stretched is lower than the position at which the intermediate transfer belt 2 is stretched in FIG. 29. Accordingly, the circumferential length of the intermediate transfer belt 2 on the primary transfer side is shorter. At this time, as illustrated in FIG. 30A and FIG. 30B, the rotation mechanism 66 further rotates counterclockwise about the rotation fulcrum 66a by the pulling force of the spring 67. Accordingly, the position at which the tension roller 65 is pressed against the intermediate transfer belt 2 changes. In other words, the tension roller 65 is pressed against the intermediate transfer belt 2 by the spring 67. Thus, the surplus of the circumferential length of the intermediate transfer belt 2 in the vicinity of the most-downstream primary transfer section 203 can be absorbed.

(144) Embodiments of the present disclosure have been described as above. However, embodiments of the present disclosure are not limited to the embodiments described above, and various modifications and improvements are possible without departing from the gist of the present disclosure.

(145) Examples of the recording sheet include, in addition to the sheet P (plain paper), thick paper, a postcard, an envelope, thin paper, coated paper such as coated paper or art paper, tracing paper, an overhead projector (OHP) sheet, a plastic film, prepreg, copper foil.

(146) In the above-described embodiments of the present disclosure, the primary transfer roller 7T and the driven rollers 33A and 21A of the most-downstream primary transfer section 203 are moved by the driving force of the common driving source. However, each of the primary transfer roller 7T and the driven rollers 33A and 21A of the most-downstream primary transfer section 203 may be moved by the driving force of a different driving source.

(147) In the above-described embodiments of the present disclosure, the distance between the primary transfer roller 7T as the most-downstream primary transfer device and the photoconductor 3T is greater at the large separation position than at the small separation position. However, the primary transfer roller 7T may not be moved when the primary transfer roller 7T is arranged at the small separation position and the large separation position.

(148) In the above description of the embodiments, the configuration in which the primary transfer rollers of all the primary transfer sections contact and separate from the corresponding one of the photoconductors has been described. However, at least any one of the primary transfer rollers of the most-downstream primary transfer section, the central primary transfer section, and the most-upstream primary transfer section, upstream from the most-downstream primary transfer section, may contact and separate from the corresponding one of the photoconductors. In addition, the transfer device does not necessarily transfer toner of five colors including a special color.

(149) Aspects of the present disclosure are, for example, as follows.

(150) First Aspect

(151) In a first aspect of the present disclosure, a transfer device includes an intermediate transferor to rotate, multiple primary transfer sections, a first tension roller, a first movement mechanism, and a second movement mechanism. The multiple primary transfer sections transfer developer images to the intermediate transferor and each of the plurality of primary transfer sections includes a primary transferor. The first tension roller is disposed downstream from a most-downstream primary transferor of a most-downstream primary transfer section most downstream among the plurality of primary transfer sections in a rotation direction of the intermediate transferor, to stretch the intermediate transferor. The first movement mechanism causes the first tension roller to move and change a position at which the tension roller stretches the intermediate transferor. The second movement mechanism causes the primary transferor of a primary transfer section upstream from the most-downstream primary transfer section in the rotation direction of the intermediate

transferor to move to a contact position at which the primary transferor contacts a latent image bearer with the intermediate transferor interposed between the primary transferor and the latent image bearer and a separation position at which the primary transferor is separated from the latent image bearer. The most-downstream primary transferor is movable between a contact position at which the most-downstream primary transferor contacts another latent image bearer with the intermediate transferor interposed between the most-downstream primary transferor and still the other latent image bearer and a separation position at which the most-downstream primary transferor is separated from the other latent image bearer. The first movement mechanism causes the first tension roller to move to at least three positions at each of which the first tension roller stretches the intermediate transferor.

(152) Second Aspect

(153) The transfer device according to the first aspect further includes at least five primary transferors, and a third movement mechanism to cause a most-upstream primary transferor, which is the primary transferor of a most-upstream primary transfer section most upstream among the plurality of primary transfer sections in the rotation direction of the intermediate transferor, to move to a contact position at which the most-upstream primary transferor contacts still another latent image bearer with the intermediate transferor interposed between the most-upstream primary transferor and the still other latent image bearer and a separation position at which the most-upstream primary transferor is separated from the still other latent image bearer.

(154) The second movement mechanism causes at least three central primary transferors, which are primary transferors of a central primary transfer section between the most-upstream primary transfer section and the most-downstream primary transfer section, to move from a contact position at which each one of the at least three central primary transferors contacts a corresponding latent image bearer with the intermediate transferor interposed between each one of the at least three central primary transferors and the corresponding latent image bearer to a separation position at which each one of the at least three central primary transferors is separated from the corresponding latent image bearer.

(155) Third Aspect

(156) The transfer device according to the second aspect further includes a first contact-and-separation mechanism to cause the most-downstream primary transferor to move to the contact position and the separation position. The first movement mechanism causes the first tension roller to move to a first position, a second position, and a third position. The first tension roller is arranged at the first position when the most-downstream primary transferor is arranged at the contact position. The first tension roller is arranged at the second position when the most-downstream primary transferor is arranged at the separation position and the primary transferor upstream from the most-downstream primary transfer section in the rotation direction of the intermediate transferor is arranged at the separation position.

(157) The first tension roller is arranged at the third position when the most-downstream primary transferor is arranged at the separation position and the primary transferor upstream from the most-downstream primary transfer section in the rotation direction of the intermediate transferor is arranged at the contact position.

(158) The first tension roller arranged at the third position is farther away from the latent image bearer than the first tension roller arranged at the second position in a direction in which the primary transferor contacts with or separates from the latent image bearer.

(159) Fourth Aspect

(160) In the transfer device according to the first or third aspect, the first contact-and-separation mechanism is the first movement mechanism.

(161) Fifth Aspect

(162) In the transfer device according to the third or fourth aspect, the second movement mechanism causes each one of the at least three central primary transferors to move from the

separation position to the contact position after the first movement mechanism has caused the first tension roller to move from the second position to the third position. The above-described movements of the at least three central primary transferors and the first tension roller correspond to the movements described in the above-described Table 1 when the mode A is switched to the mode E.

(163) Sixth Aspect

(164) In the transfer device according to the third or fourth aspect, the first movement mechanism causes the first tension roller to move from the third position to the second position after the second movement mechanism has moved each one of the at least three central primary transferors from the contact position to the separation position. The above-described movements of the first tension roller and the at least three central primary transferors correspond to the movements described in the above-described Table 1 when the mode E is switched to the mode A.

(165) Seventh Aspect

(166) In the transfer device according to the third or fourth aspect, the first movement mechanism causes the first tension roller to move from the third position to the second position after the second movement mechanism has moved each one of the at least three central primary transferors from the contact position to the separation position. The above-described movements of the first tension roller and the at least three central primary transferors correspond to the movements described in the above-described Table 1 when the mode E is switched to the mode F.

(167) Eighth Aspect

(168) In the transfer device according to the third or fourth aspect, the second movement mechanism causes each one of the at least three central primary transferors to move from the separation position to the contact position after the first movement mechanism has caused the first tension roller to move from the second position to the third position. The above-described movements of the at least three central primary transferors and the first tension roller correspond to the movements described in the above-described Table 1 when the mode F is switched to the mode E.

(169) Ninth Aspect

(170) In the transfer device according to the third or fourth aspect, the third movement mechanism causes the most-upstream primary transferor from the separation position to the contact position after the first movement mechanism has caused the first tension roller to move from the first position to the second position and the most-downstream primary transferor to move from the contact position to the separation position. The above-described movements of the most-upstream primary transferor, the first tension roller, and the most-downstream primary transferor correspond to the movements described in the above-described Table 1 when the mode B is switched to the mode F.

(171) Tenth Aspect

(172) In the transfer device according to the third or fourth aspect, the first movement mechanism causes the first tension roller to move from the second position to the first position and the most-downstream primary transferor to move from the separation position to the contact position, after the third movement mechanism has caused the most-upstream primary transferor to move from the contact position to the separation position. The above-described movements of the first tension roller, the most-downstream primary transferor, and the most-upstream primary transferor correspond to the movements described in the above-described Table 1 when the mode F is switched to the mode B.

(173) Eleventh Aspect

(174) In the transfer device according to the third or fourth aspect, the first movement mechanism causes a single driving source to move the most-downstream primary transferor and the first tension roller.

(175) Twelfth Aspect

(176) The transfer device according to any one of the second to eleventh aspects further includes a second tension roller between the most-downstream primary transferor and a primary transferor immediately upstream from the most-downstream primary transferor to stretch the intermediate transferor.

(177) The second tension roller stretches the intermediate transferor when the most-downstream primary transferor is separated from the intermediate transferor and the primary transferor immediately upstream from the most-downstream primary transferor contacts a corresponding latent image bearer.

(178) Thirteenth Aspect

(179) In the transfer device according to the twelfth aspect, the second movement mechanism causes the second tension roller to contact with and separate from the intermediate transferor.

(180) Fourteenth Aspect

(181) In the transfer device according to the twelfth aspect, the first movement mechanism causes the second tension roller to contact with and separate from the intermediate transferor.

(182) Fifteenth Aspect

(183) In the transfer device according to any one of the first to fourteenth aspects, the most-downstream primary transferor transfers developer of a special color other than any of yellow, magenta, cyan, and black to the intermediate transferor.

(184) Sixteenth Aspect

(185) In a sixteenth aspect of the present disclosure, an image forming apparatus includes the transfer device and the multiple latent image bearers.

Claims

1. A transfer device, comprising: a plurality of rollers including a first roller, a second roller and a third roller; an intermediate transferor to rotate in a rotation direction, wherein the intermediate transferor is stretched by the plurality of rollers, the first roller is upstream of the second roller and the third roller, and the second roller is upstream of the third roller; a plurality of primary transfer sections to transfer developer images to the intermediate transferor, the plurality of primary transfer sections each including a primary transferor, and the plurality of primary transfer sections are disposed between the first roller and the second roller; a tension roller which is a driven roller to stretch the intermediate transferor, the tension roller being downstream from a most-downstream primary transferor of a most-downstream primary transfer section of the plurality of primary transfer sections in the rotation direction; a first movement motor to move the tension roller and change a position at which the tension roller stretches the intermediate transferor; and a second movement motor to move a first primary transferor, of a first primary transfer section upstream from the most-downstream primary transfer section in the rotation direction, between a first contact position and a first separation position, the first contact position being at which the primary transferor contacts a latent image bearer with the intermediate transferor interposed between the first primary transferor and the latent image bearer, and the first separation position being at which the primary transferor is separated from the latent image bearer, wherein the third roller forms a secondary transfer section, the most-downstream primary transferor is movable between a second contact position and a second separation position, the second contact position being at which the most-downstream primary transferor contacts with another latent image bearer with the intermediate transferor interposed between the most-downstream primary transferor and the another latent image bearer, and the second separation position at which the most-downstream primary transferor is separated from the another latent image bearer, and the first movement motor moves the tension roller between at least three positions at each of which the tension roller stretches the intermediate transferor, the at least three positions including: a first position at which the most-downstream primary transferor is in contact with the intermediate transferor for image

transfer in a first image transfer mode, a second position at which the most-downstream primary transferor is positioned a first distance away from the intermediate transferor for image transfer in a second image transfer mode, and a third position at which the most-downstream primary transferor is positioned a second distance away from the intermediate transferor for image transfer in a third image transfer mode, the second distance being larger than the first distance.

2. The transfer device according to claim 1, further comprising at least five primary transferors; and a third movement motor to move a most-upstream primary transferor, of a most-upstream primary transfer section most-upstream among the plurality of primary transfer sections in the rotation direction, between a third contact position and a third separation position, the third contact position being at which the most-upstream primary transferor contacts still another latent image bearer with the intermediate transferor interposed between the most-upstream primary transferor and the still another latent image bearer, and the third separation position being at which the most-upstream primary transferor is separated from the still another latent image bearer, wherein the second movement motor moves at least three central primary transferors, which are primary transferors of a central primary transfer section between the most-upstream primary transfer section and the most-downstream primary transfer section, to move from contact positions to separation positions, the contact positions at which each one of the at least three central primary transferors contacts a corresponding latent image bearer with the intermediate transferor interposed between each one of the at least three central primary transferors and the corresponding latent image bearer, and the separation positions at which each one of the at least three central primary transferors is separated from the corresponding latent image bearer.

3. The transfer device according to claim 2, further comprising: a first contact-and-separation motor to move the most-downstream primary transferor between the second contact position and the second separation position.

4. The transfer device according to claim 2, wherein the first movement motor is a first contact-and-separation motor to move the most-downstream primary transferor between the second contact position and the second separation position.

5. The transfer device according to claim 4, wherein the second movement motor moves the at least three central primary transferors from the separation positions to the contact positions after the first movement motor has moved the tension roller from the second position to the third position.

6. The transfer device according to claim 5, wherein the tension roller moves when the intermediate transferor rotates.

7. The transfer device according to claim 5, wherein the another latent image bearer corresponding to the most-downstream primary transferor stops rotation during movement of the tension roller.

8. The transfer device according to claim 4, wherein the first movement motor moves the tension roller from the third position to the second position after the second movement motor has moved the at least three central primary transferors from the contact positions to the separation positions.

9. The transfer device according to claim 4, wherein the first movement motor moves the at least three central primary transferors from the separation positions to the contact positions after the second movement motor has moved the tension roller from the second position to the third position when the most-upstream primary transferor is arranged at the third contact position.

10. The transfer device according to claim 4, wherein the second movement motor moves the at least three central primary transferors from the separation positions to the contact positions after the first movement motor has moved the tension roller from the second position to the third position when the most-upstream primary transferor is arranged at the third contact position.

11. The transfer device according to claim 4, wherein the third movement motor moves the most-upstream primary transferor from the third separation position to the third contact position after the first movement motor has moved the tension roller from the first position to the second position and the most-downstream primary transferor from the second contact position to the second separation position.

12. The transfer device according to claim 11, wherein the latent image bearer other than the still another latent image bearer corresponding to the most-upstream primary transferor and the another latent image bearer corresponding to the most-downstream primary transferor stops rotation during movement of the tension roller.
 13. The transfer device according to claim 4, wherein the first movement motor moves the tension roller from the second position to the first position and the most-downstream primary transferor moves from the second separation position to the second contact position, after the third movement motor has moved the most-upstream primary transferor from the third contact position to the third separation position.
 14. The transfer device according to claim 4, further comprising a single driving source including the first motor and the second motor.
 15. The transfer device according to claim 2, further comprising: another tension roller between the most-downstream primary transferor and a primary transferor immediately upstream from the most-downstream primary transferor to stretch the intermediate transferor, wherein the another tension roller stretches the intermediate transferor when the most-downstream primary transferor is separated from the intermediate transferor and the primary transferor immediately upstream from the most-downstream primary transferor contacts a corresponding latent image bearer.
 16. The transfer device according to claim 15, wherein the second movement motor moves the another tension roller between contact with and separation from the intermediate transferor.
 17. The transfer device according to claim 15, wherein the first movement motor moves the another tension roller between contact with and separation from the intermediate transferor.
 18. The transfer device according to claim 1, wherein the most-downstream primary transferor transfers developer of a special color other than any of yellow, magenta, cyan, and black to the intermediate transferor.
 19. An image forming apparatus comprising: the transfer device according to claim 1; and a plurality of latent image bearers including the latent image bearer and the another latent image bearer.
 20. The transfer device according to claim 1, further comprising: a sensor that detects a rotation speed of the intermediate transferor, wherein the sensor moves in conjunction with movement of a corresponding primary transferor of a primary transfer section of the plurality of primary transfer sections.
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